Oral Pathology dental caries lec 1

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Dental caries

Dental caries is def ned as an irreversible, infectious, microbial disease affecting the hard tissue portion of the tooth exposed to the oral cavity, characterized by demineralization of inorganic components and destruction of organic components, resulting in cavitation.

Once penetration of the enamel has occurred, the disease progresses through the dentin to the pulp, if left untreated, the process will ultimately penetrate through the pulpal canal beyond the tooth into the marrow spaces of the bone and possibly the soft tissues, tissue spaces and muscles of the face and neck.

Causes of tooth decay

- Bacterial causes
- Non-bacterial causes (attrition, abrasion, erosion and pathological resorption).

Dental attrition is a type of tooth wear caused by tooth-to-tooth contact, resulting in loss of tooth tissue.

Abrasion is the non-carious, mechanical wear of tooth from interaction with objects other than tooth-tooth contact. It most commonly affects the premolars and canines, usually along the cervical margins.

Acid erosion is a type of tooth wear. It is defined as the irreversible loss of tooth structure due to chemical dissolution by acids not of bacterial origin.

Tooth resorption, or root resorption, is the progressive loss of dentine and cementum by the action of osteoclasts. This is a normal physiological process in the exfoliation of the primary dentition caused by osteoclast differentiation due to pressure exerted by the erupting permanent tooth. However, in the secondary dentition the process is pathological.

Factors affecting caries prevalence

Race: people living in the same geographical area but belong to different races have different caries incidence.

Age: caries increase in old and children than middle aged.

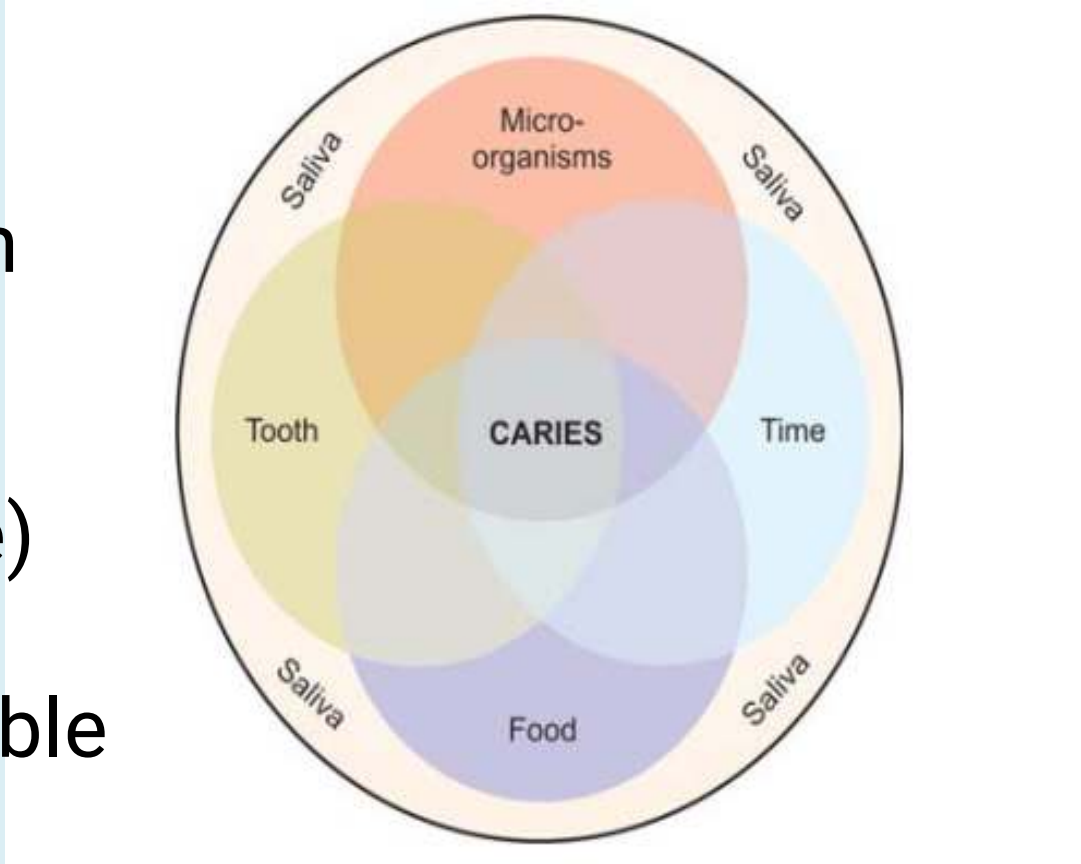
Gender: it is higher in female may be due to that their teeth erupt earlier compared to males.

Familial: heredity tendency but influenced by local factors

Requirements of Dental caries

It is multi-factorial infectious disease that requires the interaction of:

- Host (susceptible tooth surface and stagnation area)
- Bacteria (dental plaque) or bacterial plaque
- Food (dietary fermentable carbohydrates) or fermentable bacterial substrate
- Time to develop



EPIDEMIOLOGY

- The prevalence of dental caries differs among various populations throughout the world correlated with the amount of sugar consumed.
- In the more industrialized countries, where diets have traditionally had a high content of refined carbohydrates, the caries rate has been considerably higher than in less-developed countries. In recent years with the trend toward preventive measures such as fluoridated water, greater access to dental care and better oral hygiene in industrialized countries, and the concurrent rapid increase in caries activity in the less- developed societies, the large difference in caries rates has decreased.

Important measures to prevent caries:

- Extensive water fluoridation.
- Improvement in the preventive dentistry.
- Increased oral hygiene awareness.
- Increased parental care.
- More availability of skillful dental professionals.

Major theories of dental caries

- Acidogenic theory
- Proteolytic theory
- Proteolytic chelation theory
- Sucrose chelation theory
- Autoimmune theory.

Theories for the etiology of dental caries

Acidogenic theory (Miller`s chemoparasitic theory 1890):

It is the most accepted theory proposed that dental caries develop in two phases; 1st phase bacterial attack the inorganic structure to decalcified enamel and dentine by bacterial acid fermentation of dietary carbohydrates and a 2nd phase of subsequent disintegration of the organic matrix.

Proteolytic theory

The proteolytic theory of dental caries was first proposed by Gottelib in 1944, this theory states that the proteolytic enzymes liberated by cariogenic bacteria cause destruction of the organic matrix of enamel. As a result of that, the inorganic crystals of the enamel get detached from one another and finally the whole structure collapses, leading to a cavity formation.

Proteolytic- chelation theory (Schatz 1955):

The proteolytic chelation theory explains that the destruction of the organic matrix of the enamel as well as its mineral parts both occur simultaneously.

This theory suggests that firstly proteolytic break-down of the organic portion of the enamel matrix takes place. Following this, a chelating agent is formed by the combination of proteolytic breakdown products, acquired pellicle and food debris, etc which facilitates tooth decay. The whole process is helped by the bacterial enzymes which facilitates tooth decay. The chelating agent, which is formed, is always negatively charged (mostly due to its protein content) and it releases the positively charged calcium ions (Ca^{++}) from the enamel or dentin (chelation).

- **Sucrose chelation theory:** This theory suggests that ‘very high concentration of sucrose in the mouth of a caries active individual may result in the formation of complex substances like calcium saccharates and calcium complexing intermediaries, etc. by the action of phosphorelating enzymes’. These complexes cause release of the calcium and phosphorus ions from the enamel and thereby result in tooth decay.
- **Autoimmune theory:** This theory of dental caries suggests that a few odontoblast cells at some specific sites within the pulp of teeth are damaged by the autoimmune mechanisms. For this reason, the defense capacity and integrity of the overlying enamel or dentin in those specific areas are compromised, and they can be the potential sites for caries development in future.

Role of carbohydrates in dental caries

- The fermentable dietary carbohydrates play an important role in the causation of caries; the rate of caries attack depends upon the forms of taken carbohydrate and the frequency of intake.
- The risk of caries attack increases greatly if sugar is taken repeatedly in between two major meals.
- The risk of caries incidence increases greatly if the dietary sugar is sticky in nature which can remain adhered to the tooth surfaces for a longtime after taking the meal.

- The glucose, sucrose and fructose, etc. are rapidly diffused into the plaque due to their low molecular weight and available for fermentation by bacteria.
- Starches produce little or no caries because they are very slowly diffused into the plaque and they also require extra cellular amylase to hydrolyze.
- Nonsugar sweeteners, e.g. saccharin, aspartine and thaumatin, etc. are non-cariogenic.

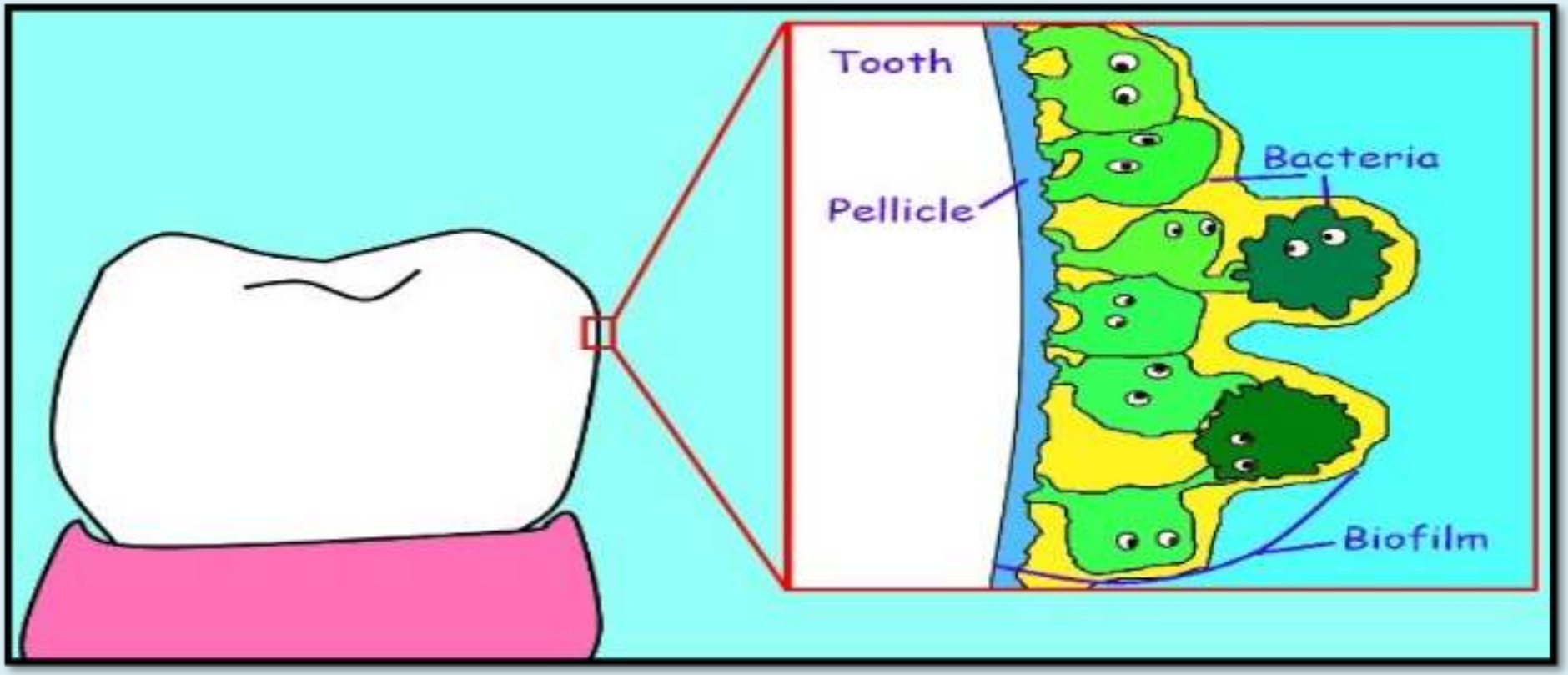
Following the ingestion of these sugars, the pH of the plaque falls to 4.5 to 5 within 1 to 3 minutes and it takes another 10 to 30 minutes to return to neutrality. The critical pH of plaque is 5.5 and tooth demineralization starts if the pH drops below this level.

Role of bacterial plaque in dental caries

- Plaque is a thin, transparent film produced on the tooth surface and it consists of food debris, desquamated epithelial cells, leukocytes, microorganisms suspended in salivary mucins and extracellular bacterial polysaccharides (glucans).
- Acquired pellicle is a component of the dental plaque, which is made by the salivary glycoprotein and is formed just prior to the bacterial colonization.

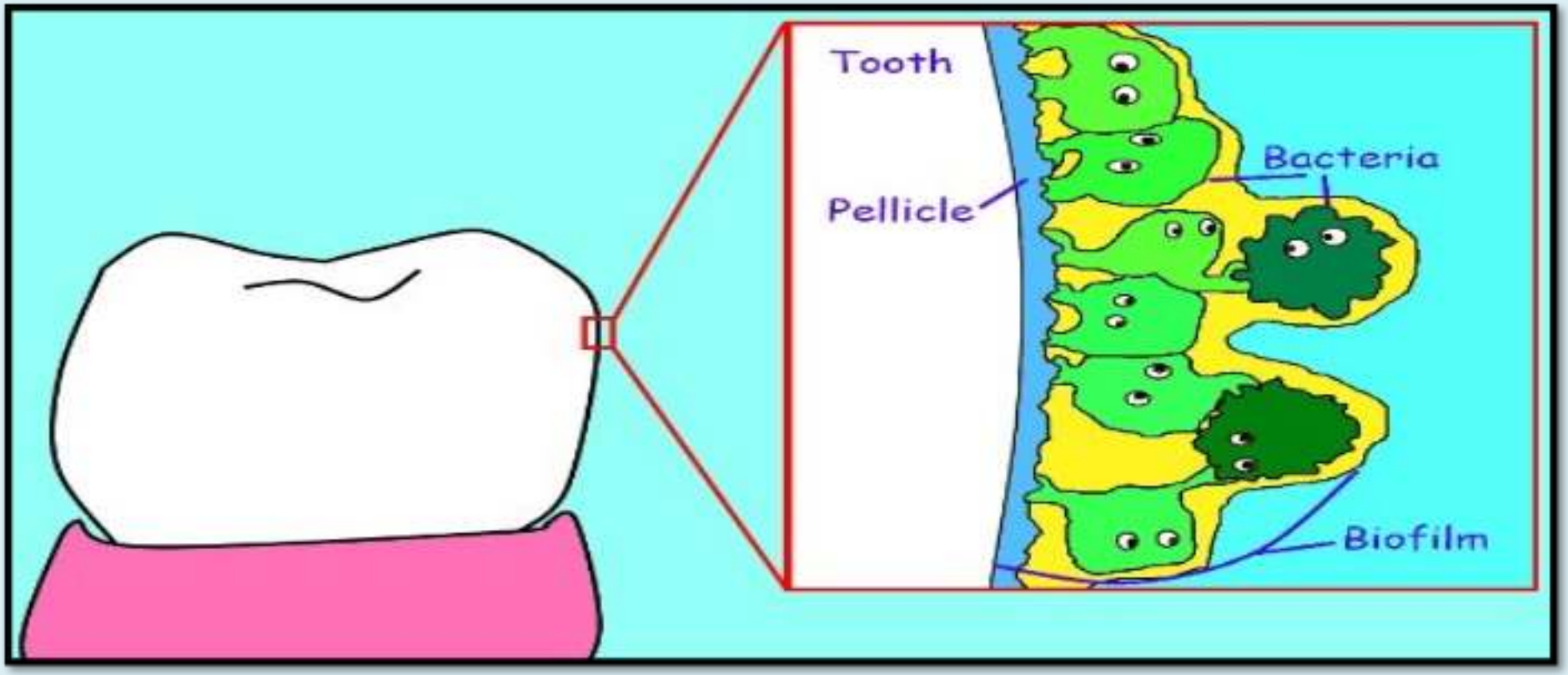
One hour after the formation of acquired pellicle, some organisms such as S. sanguis , A. viscous etc. become attached to it. These organisms are called ‘pioneering organisms’ in dental caries. These initial organisms lack in caries producing potential since they are mostly aerobic in nature and produce very little amount of acid by the fermentation of the carbohydrates.

As the plaque matures with time, S. mutans group becomes more predominant within the plaque. These organisms rapidly metabolize the carbohydrates and produce organic acids.



Role of microorganisms in dental caries

A large number of microorganisms play roles in the development of dental caries and among them the most important one is the Streptococcus mutans, Actinomyces israelii and Lactobacillus organisms were considered to be important organisms in the progression of caries lesion.



Contributing factors in dental caries

- Intrinsic factors: Tooth factor
- Extrinsic factors: Saliva factor, Diet factor, Systemic factors and Immunity
- Tooth factor: Not all the teeth in the dental arch or all surfaces of the individual tooth are equally susceptible to caries. Composition of Tooth, permeability of the enamel surface, morphology and position of tooth all are influence the process of dental caries.
- Saliva factor: saliva plays a very important role in the prevention of dental caries, increasing the flow rate, viscosity, buffering capacity

DIET FACTOR

- Physical Nature of Diet; If the diet contains sufficient amount of fibrous foods that help to keep the teeth clean as well as stimulates the salivary flow, the chances of caries formation will be less. Whereas more and more intake of soft and sticky foods increases the possibility of caries development.
- Composition of the Diet; Presence of phosphates, Traces of molybdenum and vanadium in the diet may reduce the incidence of caries. The diet that contains adequate amounts of vegetables, vitamin (A, D, K and B-complex) and minerals, etc. is often associated with a low caries incidence.

- **Systemic factor:** Some people hereditarily have an increased tendency to develop caries while other people show just the reverse tendency.
- **Immunity:** Immune mechanism plays an important role in the prevention of caries in humans including serum and salivary antibodies IgG, IgA and IgM clones as well as cell-mediated immune response. The humoral and cell-mediated immune systems gain access into the oral cavity either via the gingival cervicular fluid or via the saliva.

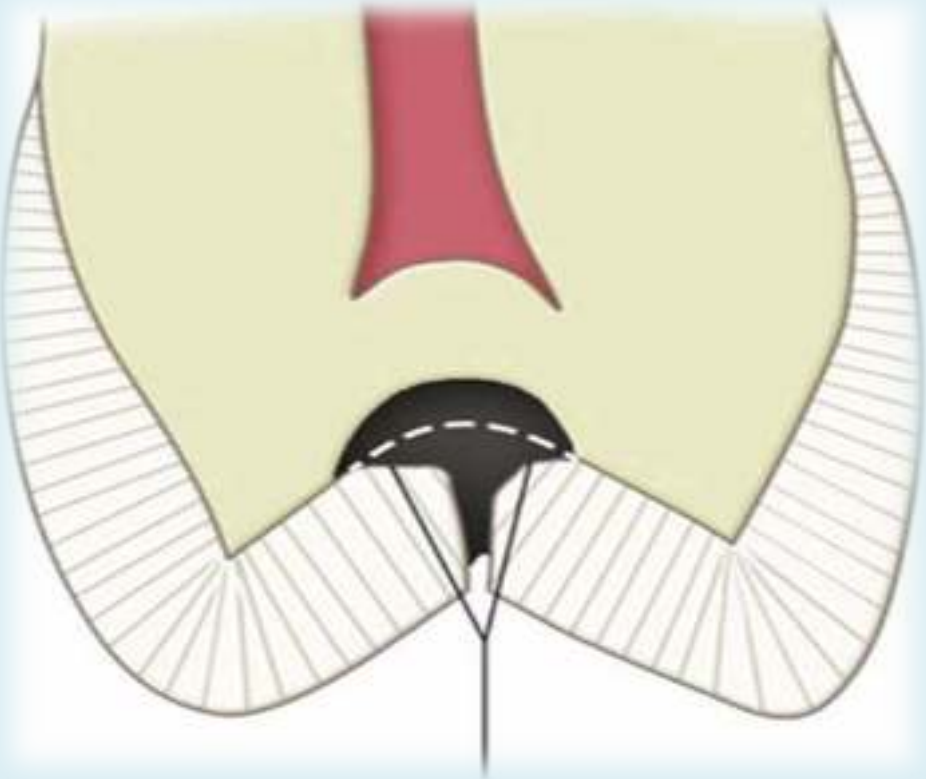
CLINICAL TYPES

1. Pit and fissure caries:

This type of lesion occurs in occlusal surfaces of molars and premolars, buccal and lingual surfaces of molars and lingual surfaces of maxillary incisors.

The lesions usually appear brown or black, when the lesion is examined by a fine explorer tip, a “catch point” is often felt.

The lesions are smaller in the beginning but become wider as they spread towards the dentin due to the typical orientation of the enamel rods.



When the lesions reach the dentinoenamel junction (DEJ), they spread laterally to cause undermining of the enamel. The enamel directly bordering the pit or fissure may appear opaque and bluish-white as it becomes undermined and damage by force of mastication.

2. Incipient Caries:

Initial carious lesion limited to the enamel, intact enamel surface but there is destruction of the enamel below the surface layer “subsurface demineralization”.

“chalky-white” appearance of the tooth surface only of the tooth is dry and the typical chalky-white condition disappears if the surface of the tooth becomes wet. The incipient caries is a reversible process and the lesion can be cured due to remineralization by salivary mineral ions and caries can be prevented by topical fluorides.

3. Smooth Surface Caries:

This type of carious lesion occurs in relation to the smooth surfaces of teeth, e.g. proximal surfaces or gingival areas of the buccal and lingual aspect of tooth. The lesion beings as chalky-white opacity of enamel with progress it becomes pigmented brown or bluish.

4. Rampant Caries:

This is an acute fulminating type of carious lesions involved multiple number of teeth (may be all teeth) in multiple surfaces. Rapid coronal destruction occurs within a short span of time, causing early involvement of the pulp. The common age of occurrence of rampant caries is about 4 to 8 years for the deciduous teeth and 11 to 19 years for the permanent teeth.

5. Nursing Bottle Caries:

It is an acute carious lesion occurs among children who take milk or fruit juices by the nursing bottle, for a considerably longer duration of time, preferably during sleep. Nursing bottle caries commonly occurs in the upper anterior teeth (as these are constantly coming in contact with the sweetened milk); while the lower teeth are not usually affected as they remain under the cover of the tongue.

Both the nursing bottle caries and rampant caries cause early pulp involvement because they spread at a very rapid pace and as a result, the pulp hardly gets any time to protect itself by forming reparative dentin.

6. Chronic Caries:

This type of caries progresses slowly and it rarely causes pulp involvement (unless the tooth is left untreated for many years) because the pulp gets suf ficient time to produce secondary dentin or reactionary dentin to protect itself.

7. Arrested Caries:

- It is a ceased carious lesion after the initial development. It can occur both in enamel and in dentin. This process occurs due to remineralization from saliva or due to topical fluoride applications, or gets exposed to the cleaning measures like tooth brushing and salivary secretions.
- The arrested caries presents a hard, black or brown-colored dentinal surface that is highly mineralized due to remineralization from oral fluids and has increased fluoride content.

8. Recurrent Caries:

A carious lesion that begins around the margins or at the base of a pre-existing defective restoration.

9. Forward Caries: Carious lesion progresses unidirectionally from enamel into the dentin and pulp.

10.Backward Caries: Carious lesion also initially progress from enamel into the dentin, where they spread laterally and involve a wide area. Later on, these lesions proceed in a backward direction from dentin back to enamel and affect the enamel once again at a different location.

11. Root Caries (cemental caries):

These are carious lesions that involve the cemental wall of the exposed root surfaces of teeth (because of aging or gingival recession), the plaque accumulates on the rough surface of exposed cementum and progress rapidly into the dentine and pulp, the actinomycotic groups of organisms are mostly responsible for the causation of root caries.

12. Radiation Caries:

Patients receiving large doses of radiation for the treatment of malignant lesions in the head and neck region, often develop a specif ċ type of large caries lesions in the cervical areas of the teeth. The exact cause of radiation caries is not known, but it may be due to the reduced salivary secretions, secondary to the radiotherapy.

Histopathology of dental caries

Enamel caries

In enamel, caries process may start in the pits and fissures or in the smooth surface. Caries in pit and fissures spread in a triangular pattern with the base towards the dentinoenamel junction and apex towards the surface. In smooth surface caries although the spread follows a triangular pattern the base is towards the surface of the enamel and the apex towards the dentinoenamel junction.

Histopathology:

1. Translucent zone: it is the deepest zone in the caries enamel lesion, it is more porous than intact enamel and dissolution of mineral occurs mainly at the junction of prismitic and interprismitic enamel.

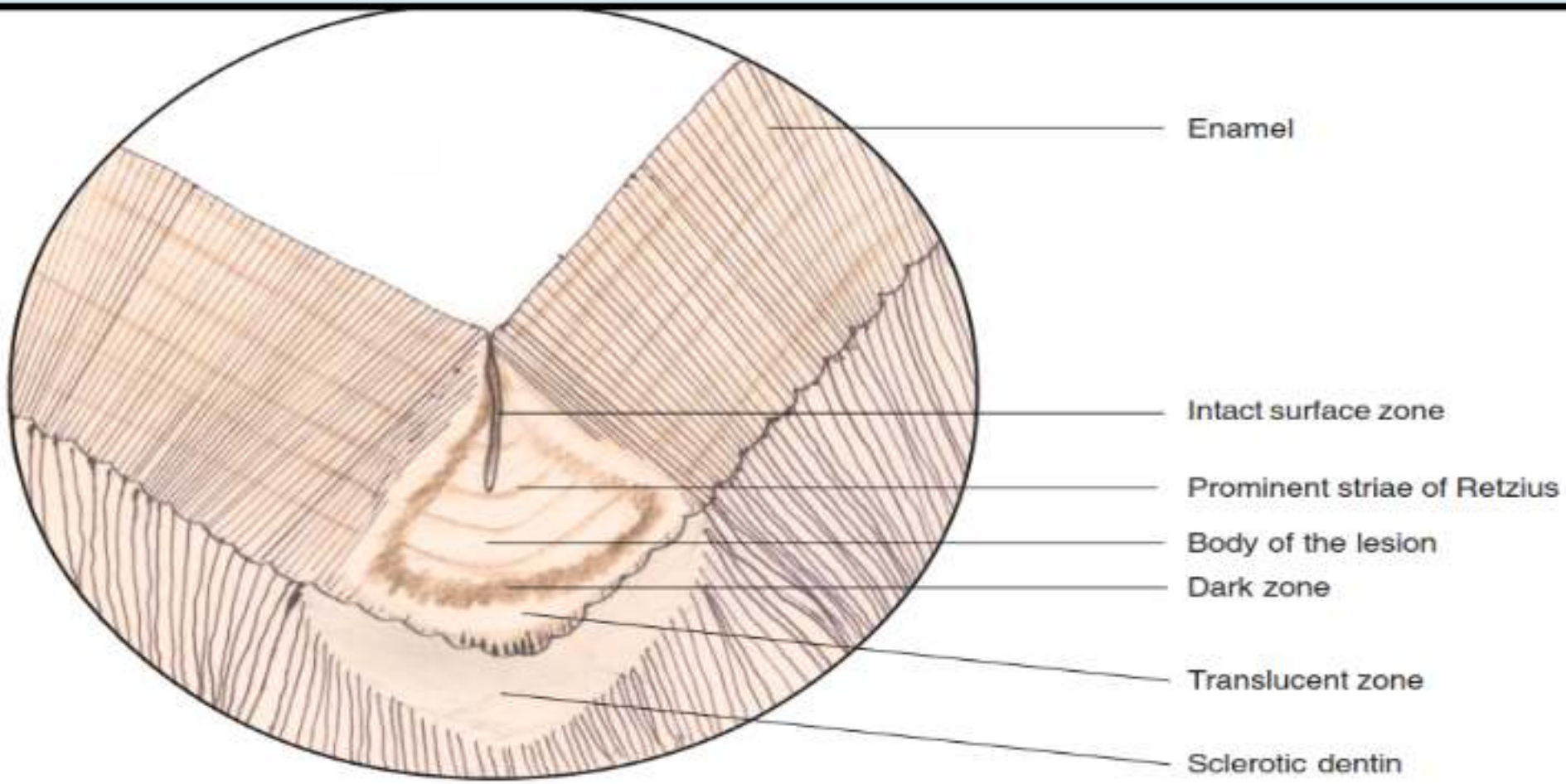


Fig. 14.1: Enamel caries (pit and fissure—ground section)

2. Dark zone: this zone is located superficical to the translucent zone and its dark appearance is due to the excessive demineralization of the enamel, This zone is narrower in rapidly advancing caries and it is wider in slowly advancing lesions. It contains more pores than the translucent zone .This zone also reveals some degrees of remineralization of the carious enamel.

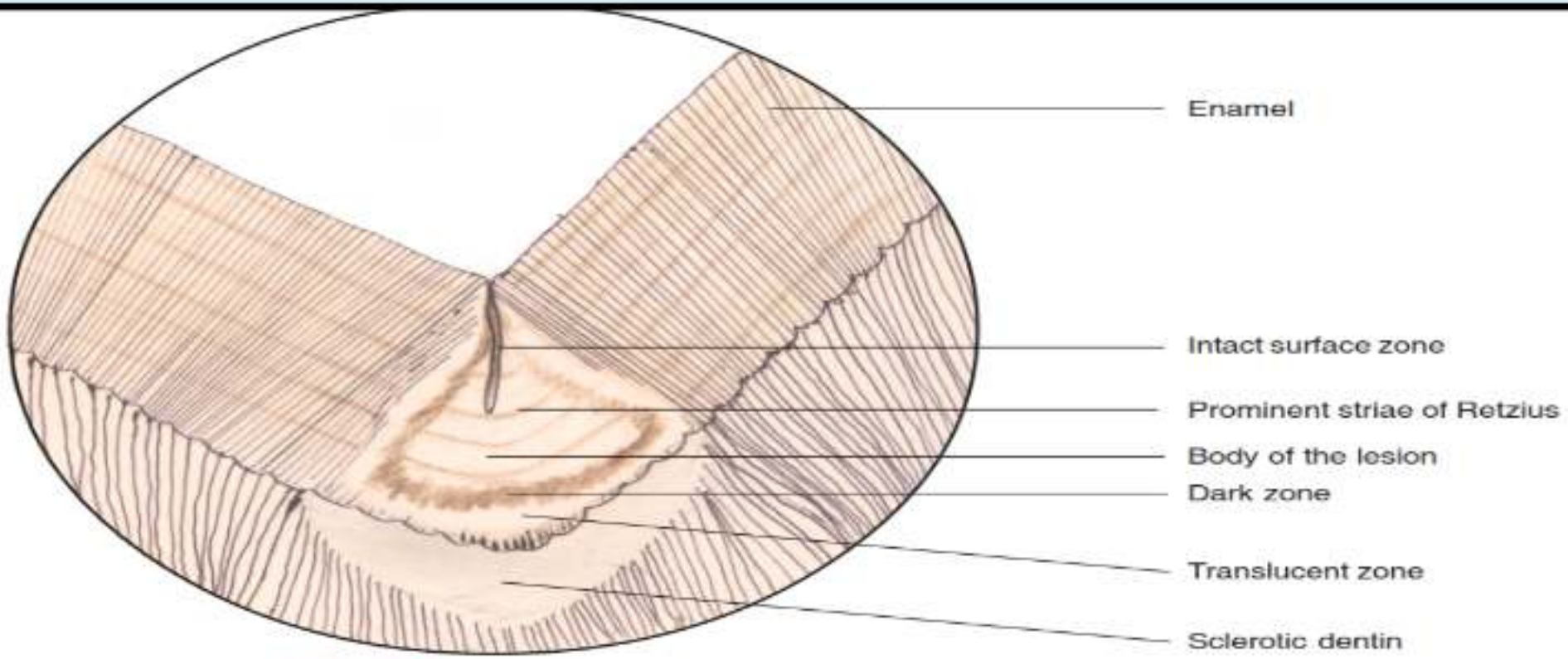


Fig. 14.1: Enamel caries (pit and fissure—ground section)

3. Body of the lesion: This zone is situated between the dark zone and the surface layer of enamel, and it represents the area of greatest demineralization. It has a more pores than the dark zone, in this zone striae of Retzius appear more prominent.

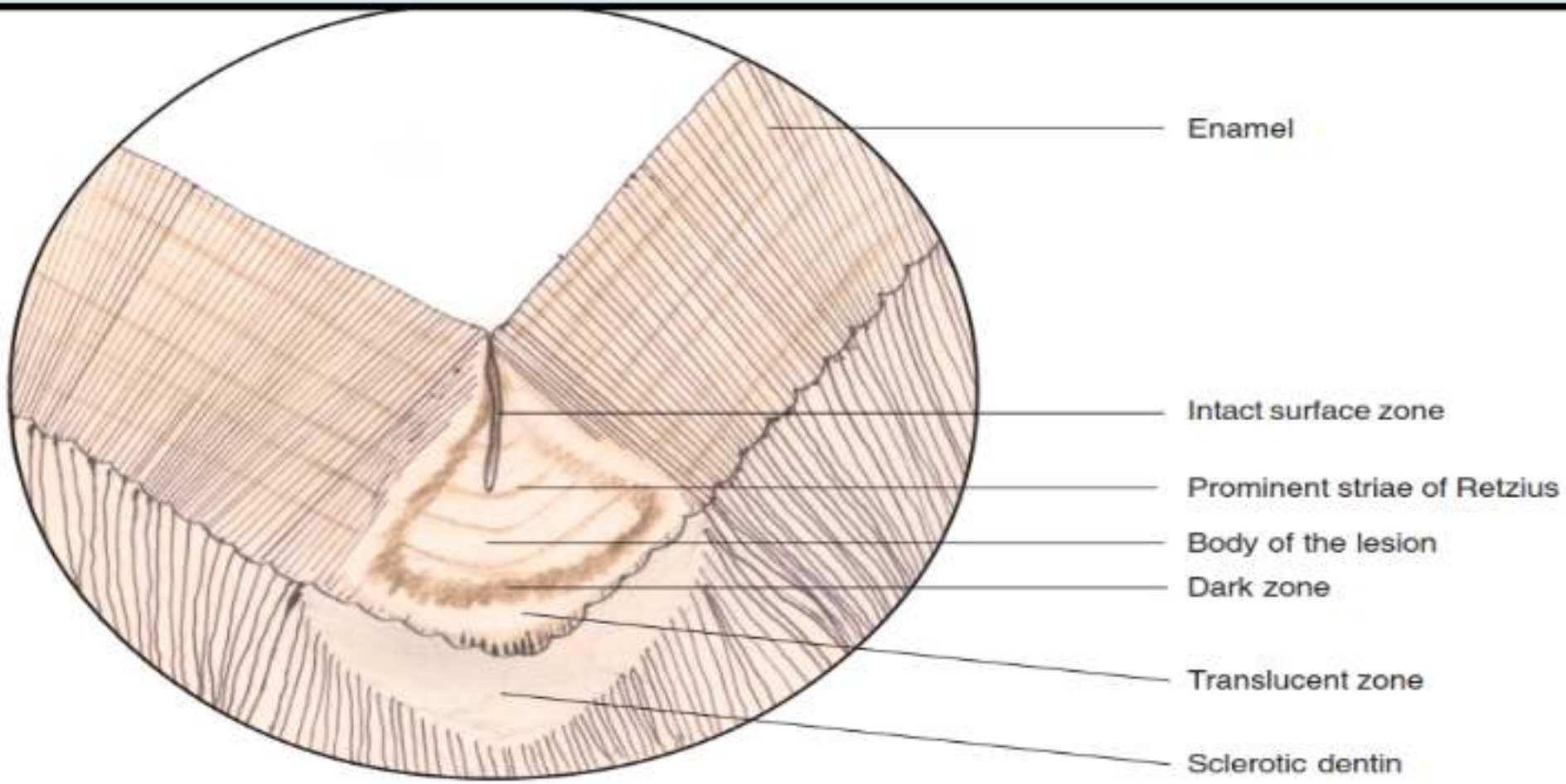


Fig. 14.1: Enamel caries (pit and fissure—ground section)

4. Surface zone: This is about 40 microns thickness at the surface of enamel caries lesion. Initially, the surface zone of a carious enamel remains unaffected despite subsurface demineralization and this due to surface remineralization results from active precipitation of mineral ions derived from both plaque and the saliva. However, in untreated cases the surface enamel often gets destroyed and a cavity is formed.

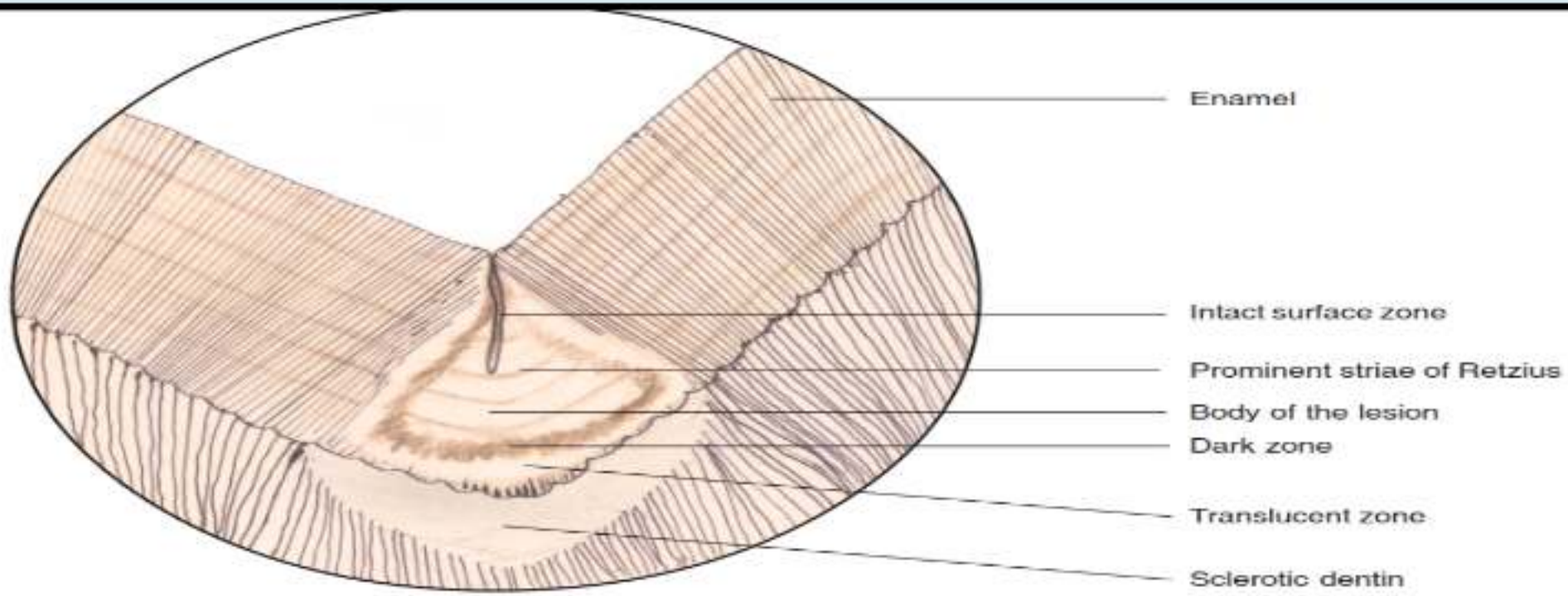


Fig. 14.1: Enamel caries (pit and fissure—ground section)

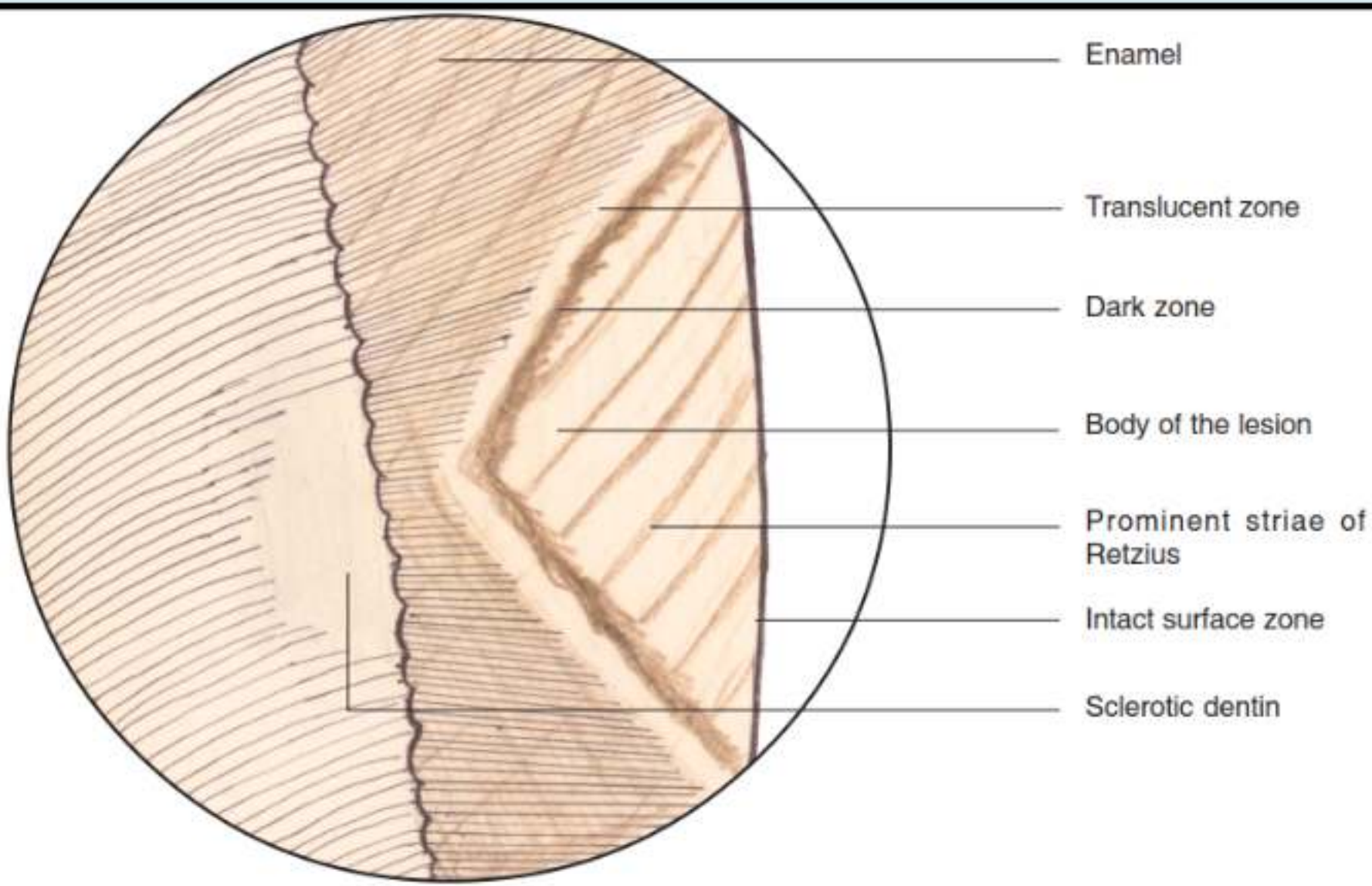


Fig. 14. 2: Enamel caries (smooth surface—ground section)

Dentinal caries histopathology

Once the caries process spreads to dentin, the microorganisms invade the dentinal tubules and release acids. The spread of caries follows a triangular pattern with the base towards the dentino-enamel junction and apex towards the pulp. Apex will be located more apical to the base.

Histopathology: Microscopically five different zones are seen:

- **Zone I (Normal Dentin):** This zone represents the innermost layer of the carious dentin and here the dentinal tubules appear normal. No bacteria in the tubules and no crystal deposit in the tubules.

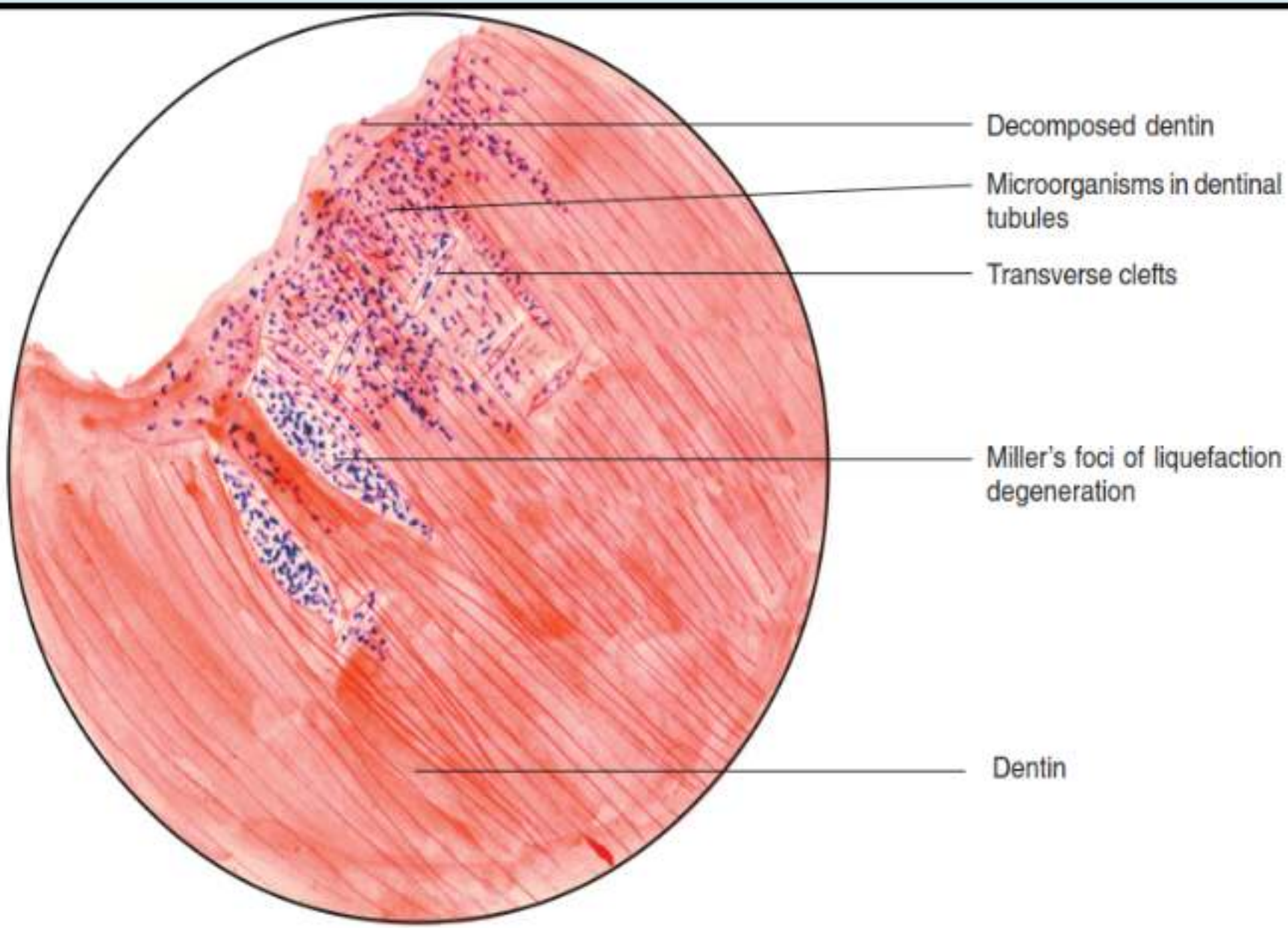


Fig. 14.3: Dentinal caries (decalcified section)

- **Zone II (Sub-transparent Dentin):** This is the zone of dentinal sclerosis and it is characterized by the deposition of very fine crystal structures within the dentinal tubules. The superficial layer shows areas of demineralization and damage of the odontoblastic processes. No bacteria in the tubules. This dentin is capable of remineralization.

- **Zone III (Transparent Dentin):** This zone appears transparent and this is because of the demineralization of dentin due to caries. It is softer than normal dentin. Further loss of mineral ions from intertubular dentin. Large crystals within the lumen of the dentinal tubules. No bacteria in tubules. This dentin is capable of remineralization.

- **Zone IV (Turbid Dentin):** This zone is called the “turbid dentin” and is marked by the widening and distortion of the dentinal tubules, which are packed with microorganisms. There is very little amount of mineral present in the dentin and moreover denaturation of collagen fibers also takes place. This zone cannot undergo self-repair or remineralization. Must be removed before restorative treatment.

- **Zone V (Infected Dentin):** it is the outermost zone of the carious dentin and is characterized by complete destruction of the dentinal tubules (are called the liquefaction foci of Miller). Lateral spread of organisms through the lateral branches of dentinal tubules results in formation of ‘transverse clefts’ which are perpendicular to the tubules. Accumulation of a large number of microorganisms and their by-products that invade and destroy the peri- and intertubular dentin results in destruction of entire dentinal structure.

Protective responses of dentin and pulp against caries

Dentin is a living tissue containing odontoblastic cells that react in response to stimulus (bacterial invasion) in the dentin-pulp complex in order to protect the tooth from the invasion of caries. Following are the protective responses from the pulp-dentin complex:

- **Tubular sclerosis:** Peritubular dentin reduces the size of the individual dentinal tubules and thus prevents the bacterial penetration.
- **Irregular reactionary dentin (tertiary dentin):** This type of dentin forms under moderate to severe insult by caries and this irregular reactionary dentin contains only few irregularly shaped and tortuous tubules. Sometimes, the dentinal tubules can be even absent.
- **Dead tracts:** The dead tracts form when the odontoblast cells die and their tubules become sealed off. They often prevent the further progression of caries in dentin towards the pulp.

Regular reactionary dentin (secondary dentin):
Reactionary dentin develops on regular basis as a result of localized, nonspecific mild irritation to the odontoblast cells. However, this dentin seems to develop mostly during caries attack in a tooth. In regular reactionary dentin a layer of dentin forms at the surface of the pulp chamber deep to the progressive front of dentin caries and it contains normal tubular structures. It is often hypermineralized as compared to the primary dentin and it delays the pulp exposure by increasing the depth of the tissue between carious dentin and the pulp.

Methods of caries prevention

- **Limit substrate:** eliminate sucrose from the diet or reduce its amount and eliminate sucrose from the "in-between" meals and snacks.
- **Modify oral microflora:** Bactericidal mouth-rinse by chlorhexidine, Topical fluoride treatments, Antibiotic treatment by vancomycin and tetracyclines
- **Plaque disruption**—by brushing and flossing.
- **Modify tooth:** Systemic fluorides, Topical fluorides, Maintain a smooth surface of the tooth, Stimulate salivary flow, Eat noncariogenic fibrous foods that require lots of chewing and Use sugarless chewing gums.
- **Restore tooth surface:** Restore all cavitated lesions, Seal pits and fissures at caries risk, Correct all defects, e.g. marginal crevice, proximal overhangs.

Reactionary (or tertiary)dentine

It is a layer of dentine formed as a non-specific response of irritated odontoblasts to caries, tooth wear, cavity and crown preparations.

Charactization of reactionary dentine: Irregular, Tortuous, Variations in mineralization, it is hypermineralized comparing with primary dentine.

Clinical aspects of dentine caries

The soft stained or unstained dentine at the base of a cavity should be removed with no need to remove hard stained dentine that is not have catch with probe or we used stepwise excavation of a deep carious lesion to remove infected dentine without pulpal exposure (sealing interval from 4 to 6 months) to allow time for the reactive dentin to develop.

Immunological aspects of dental caries

- Humoral response against *S. mutans* is provided largely by secretory IgA antibodies in saliva which interfere the attachment of the organism to tooth surfaces, IgG and IgM antibodies as well as cell-mediated immune response can also gain access to the mouth via the crevicular fluid.
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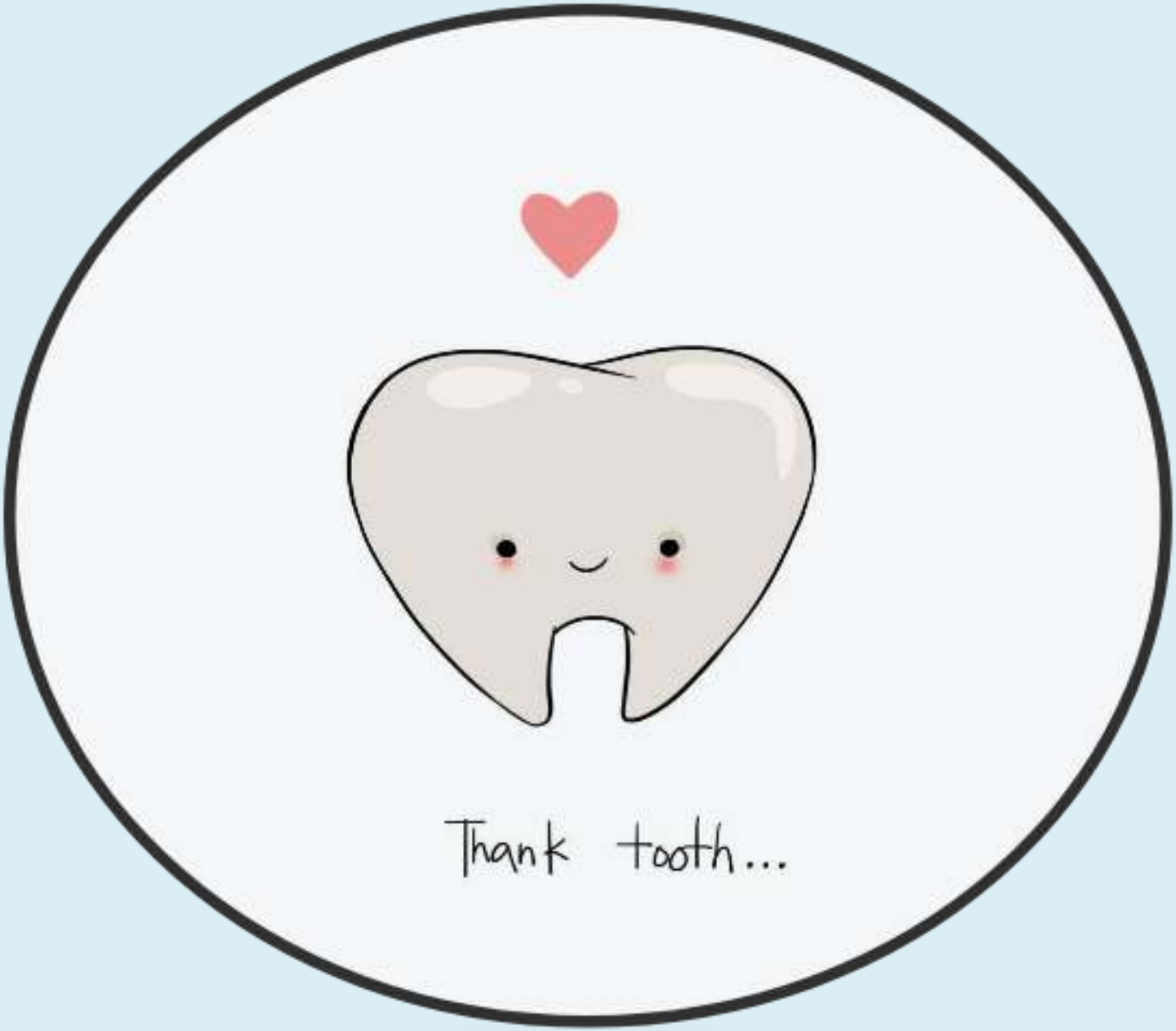
Thank tooth...